

Collinear and coplanar vectors

Two tests, both from the definition: one scalar for collinearity, two for coplanarity.

LESSON 6

1 × 40 MIN

GEOMETRY 11, §3

P1 · VECTORS

LESSON CLOCK

8'

12'

14'

8'

3'

Collinearity	8 min
Proving three points collinear	12 min
Coplanarity	14 min
Practice	8 min
Homework	3 min

LEARNING OBJECTIVES

- Test two vectors for parallelism.
- Prove three points are collinear using vectors.
- Express a vector as a linear combination of two others.
- Decide whether three vectors are coplanar.

TEXTBOOK

National	pp. 29–34
Cambridge	Core & Extended, pp. 335–338
Workbook	Exercise 3.1

01

Explanation

Collinear vectors

THE TEST

a and b are **collinear** (parallel) exactly when $b = ka$ for some scalar k — equivalently, when their components are in a constant ratio: $\frac{b_1}{a_1} = \frac{b_2}{a_2} = \frac{b_3}{a_3}$.

$(2, -4, 6)$ and $(1, -2, 3)$ are collinear with $k = 2$. $(1, 2, 3)$ and $(2, 4, 5)$ are not: the first two ratios give 2 but the third gives $\frac{5}{3}$.

ALL THREE RATIOS, NOT TWO

Two agreeing ratios prove nothing. This is exactly how a “parallel” answer goes wrong.

Three points collinear

Points A, B, C lie on one line exactly when AB and AC are collinear vectors — they already share the point A , so a common direction is enough.

$$A(1,2,3), B(3,5,7), C(7,11,15): AB = (2,3,4), AC = (6,9,12) = 3AB$$

So the three are collinear, and $AB : BC = 1 : 2$.

Coplanar vectors

THE TEST

Three vectors are **coplanar** when one is a linear combination of the other two: $c = \alpha a + \beta b$. Geometrically, all three can be drawn in one plane.

Method. Write the equation componentwise, giving three equations in α and β . Solve two of them and check the third. If it holds, they are coplanar; if it fails, they are not.

$$c = \alpha a + \beta b \Rightarrow \text{three equations, two unknowns} \Rightarrow \text{check for consistency}$$

Two vectors are always coplanar — two directions determine a plane. Coplanarity only becomes a real question with three.

02 Worked examples

EXAMPLE 1 Are $a = (2, -6, 4)$ and $b = (-3, 9, -6)$ collinear?

$$\textcircled{1} \quad \frac{-3}{2} = -1.5$$

$$\textcircled{2} \quad \frac{9}{-6} = -1.5$$

$$\textcircled{3} \quad \frac{-6}{4} = -1.5$$

All three agree.

ANSWER

Yes, with $b = -1.5a$

EXAMPLE 2 Are $A(1,0,2)$, $B(3,2,4)$, $C(6,5,7)$ collinear?

① $AB = (2,2,2)$

② $AC = (5,5,5)$

③ $AC = 2.5 AB$

ANSWER

Yes

EXAMPLE 3 Are $a = (1,0,1)$, $b = (0,1,1)$, $c = (2,3,5)$ coplanar?

① $c = \alpha a + \beta b$

② $\alpha = 2$ from the first component.

③ $\beta = 3$ from the second.

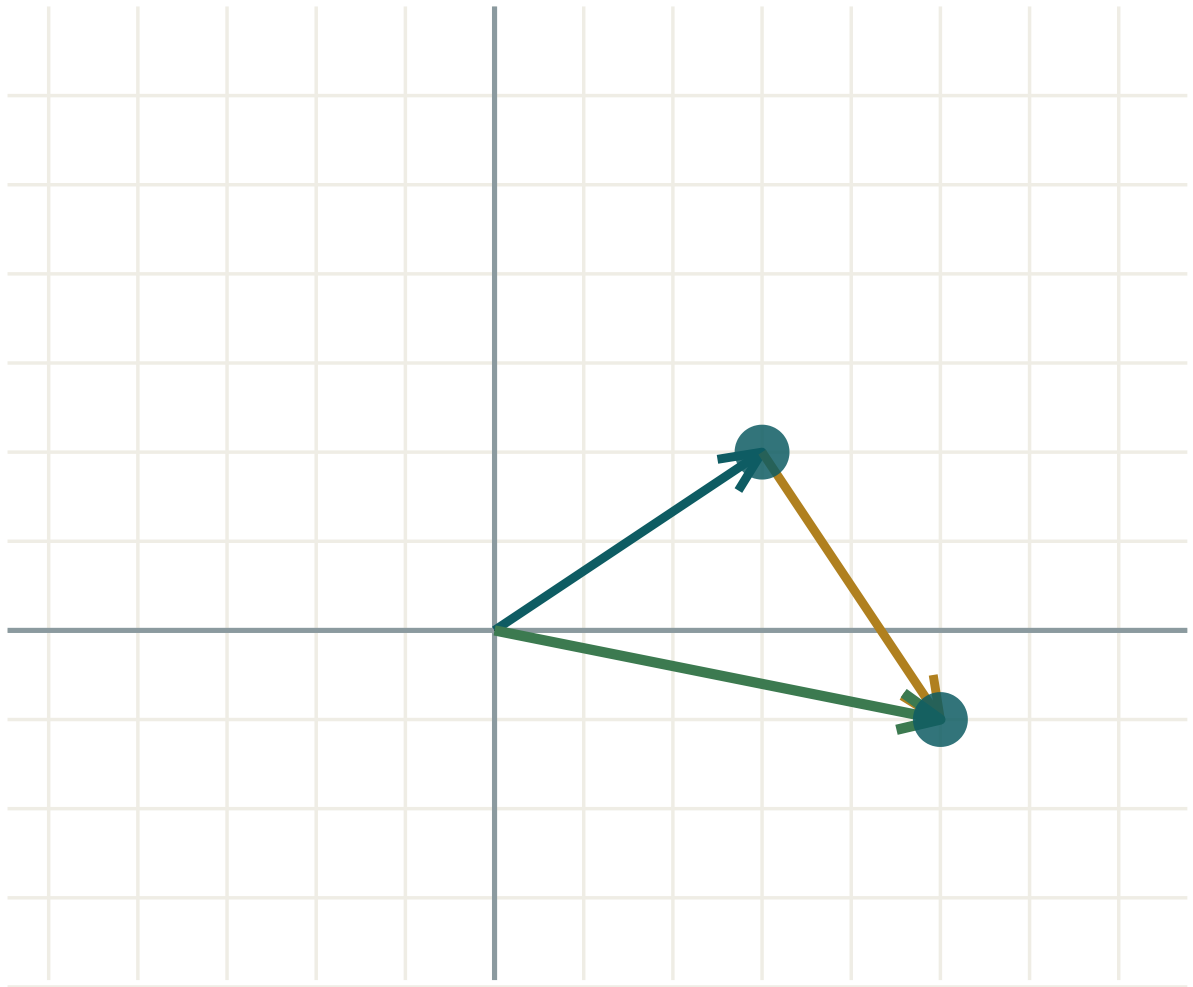
④ Third: $\alpha + \beta = 5$ ✓

ANSWER

Yes — $c = 2a + 3b$ **03 Interactive model**

Draw two vectors, then a third, and ask whether a sheet of paper can hold all three.

Combining vectors



$a \ (3; \ 2)$

$b \ (2; \ -3)$

$a + b \ (5; \ -1)$

$|a| \ 3.61$

$|b| \ 3.61$

$|a + b| \ 5.1$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Collinear vectors	Kollinear vektorlar	Коллинеарные векторы
Coplanar vectors	Komplanar vektorlar	Компланарные векторы
Linear combination	Chiziqli kombinatsiya	Линейная комбинация
Parallel vectors	Parallel vektorlar	Параллельные векторы
Decomposition	Yoyish	Разложение
Linearly independent	Chiziqli erkli	Линейно независимые
Scalar coefficient	Skalyar koeffitsient	Скалярный коэффициент
Common point	Umumiy nuqta	Общая точка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

a and b are collinear when:

$(1,2,3)$ and $(2,4,6)$ are:

Three vectors are coplanar when:

Two vectors are:

never coplanar

always coplanar

sometimes coplanar

always collinear

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Are $(1,2,3)$ and $(3,6,9)$ collinear?

ANSWER yes

2. Are $(1,2,3)$ and $(2,4,7)$ collinear?

ANSWER no

3. Find k : $(4,8,12) = k(1,2,3)$

ANSWER $k = 4$

4. Are $(0,0,1)$ and $(0,0,-5)$ collinear?

ANSWER yes

5. Are two vectors always coplanar?

ANSWER yes

6. Is $(2,2,2)$ parallel to $(1,1,1)$?

ANSWER yes

7. $AB = (1,1,1)$, $AC = (3,3,3)$. Are the points collinear?

ANSWER yes

Medium · the standard question

7 PROBLEMS

1. Are $A(1,0,2)$, $B(3,2,4)$, $C(6,5,7)$ collinear?

ANSWER **yes**

2. Are $A(0,0,0)$, $B(1,1,1)$, $C(2,2,3)$ collinear?

ANSWER **no**

3. Find t so $(2, t, 6)$ is parallel to $(1, 3, 3)$

ANSWER $t = 6$

4. Is $(2,3,5)$ in the plane of $(1,0,1)$ and $(0,1,1)$?

ANSWER **yes**

5. Express $(3,4,7)$ from $(1,0,1)$ and $(0,1,1)$

ANSWER $3a + 4b$

6. Are $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ coplanar?

ANSWER **no**

7. Find the ratio $AB : BC$ for $A(1,2,3)$, $B(3,5,7)$, $C(7,11,15)$

ANSWER $1 : 2$

Hard · stretch and reason

7 PROBLEMS

1. Are $(1,2,1)$, $(2,1,3)$, $(4,5,5)$ coplanar?

ANSWER yes — $c = 2a + b$

2. Are $(1,1,0)$, $(0,1,1)$, $(1,0,2)$ coplanar?

ANSWER no

3. Find λ so $(1,2,\lambda)$ is coplanar with $(1,0,1)$ and $(0,1,1)$

ANSWER $\lambda = 3$

4. Prove the midpoints of the sides of a space quadrilateral form a parallelogram

ANSWER Each midline equals $\frac{1}{2}$ of a diagonal

5. Prove the medians of a triangle are concurrent using vectors

ANSWER All meet at $\frac{a+b+c}{3}$

6. Show that if $c = \alpha a + \beta b$ then a , b , c cannot span space

ANSWER They lie in the plane of a and b

7. Find P on AB with $AP : PB = 2:3$ for $A(1,1,1)$, $B(6,11,16)$

ANSWER $(3, 5, 7)$

07 Homework

Homework — 4 tasks

Check all three ratios, never two.

- Are $(3, -6, 9)$ and $(-2, 4, -6)$ collinear? Show the ratios.
- Are $A(2,1,0)$, $B(4,4,3)$, $C(8,10,9)$ collinear? Give $AB : BC$.
- Show that $(3, 5, 8)$ is coplanar with $(1, 1, 2)$ and $(1, 3, 4)$.
- Find λ so that $(2, \lambda, 5)$ is coplanar with $(1, 0, 1)$ and $(0, 1, 2)$.